

Expedia Hotel Recommendations

Evros Vasileiou
Computer Science
University of Nicosia
Nicosia, Cyprus
vasileiou.e8@live.unic.ac.cy

ABSTRACT

Online travel platforms rely heavily on recommendation systems to help users navigate large numbers of available hotels. This project presents a session-based recommendation approach using the Expedia dataset, where user interactions are modeled as sequences within browsing sessions. To capture relationships between hotels, Word2Vec is applied to learn dense vector representations by treating sessions as sentences and hotels as words. Three recommendation strategies are evaluated: a popularity-based baseline, a previous-item model, and a context-based model using session embeddings. Performance is assessed using the Hits@10 metric. The results suggest that embedding-based approaches outperform the baseline, with the previous-item model achieving the best performance. These findings suggest that sequential patterns play an important role in modeling user behavior. Overall, the study indicates that relatively simple embedding techniques can effectively capture user behavior patterns in large-scale datasets, even without explicit user profile information.

1 Introduction

Modern travel platforms must assist users in selecting relevant hotels from an overwhelming number of options. Traditional recommendation approaches based solely on popularity fail to account for the dynamic nature of user behavior. In practice, users interact with multiple hotels within a single session, forming meaningful sequences of exploration. These interaction sequences reflect evolving user behavior, where each click provides additional context about the user's current browsing patterns.

This work formulates hotel recommendation as a session-based next-item prediction problem, where the objective is to predict the next hotel a user will select based on prior interactions within the same session. To address this, the project employs Word2Vec to learn representations of hotels from sequential data. The effectiveness of three approaches is examined: a frequency-based baseline, a model leveraging the most recent interaction, and a method incorporating the full session context. This formulation aligns with practical recommendation scenarios where models can benefit from adapting to ongoing user behavior.

2 Related Work

Recommendation systems are typically divided into collaborative filtering, content-based, and hybrid methods. While collaborative filtering relies on user-item interactions, it often struggles with sparse or rapidly changing data. Content-based methods depend on item attributes but may overlook implicit behavioral patterns.

Hybrid approaches attempt to combine these methods but often require additional metadata that may not always be available. Session-based recommendation has emerged as a suitable approach when only short-term user activity is available. Methods such as Markov chains, nearest-neighbor techniques, and recurrent neural networks have been widely explored. More recently, embedding techniques inspired by natural language processing have gained attention. Word2Vec learns item representations based on co-occurrence patterns and provides a simpler alternative to more complex neural architectures. Its ability to capture latent relationships between items makes it useful in recommendation tasks involving large interaction datasets. Recent research has also explored graph-based methods, where items are represented as nodes and user transitions as edges, enabling the modeling of complex relationships between items. These approaches further highlight the importance of structural information in sequential recommendation tasks.

3 Dataset and Preprocessing

The dataset contains approximately 12,709,329 sessions, 79,102,862 total interactions, and 1,166,727 unique hotels, where each session is represented as a sequence of hotel identifiers. The raw data includes a column containing comma-separated hotel IDs, which are transformed into ordered lists for modeling purposes. This transformation preserves the temporal order of interactions, which is essential for sequence-based modeling.

Sessions with fewer than two interactions are excluded, as they do not provide sufficient information for prediction. A session length feature is computed to analyze user behavior. The dataset contains a large number of sessions and a high number of unique hotels, highlighting the scale and sparsity of the problem. Such sparsity introduces challenges for learning meaningful representations, particularly for rarely occurring items.

To ensure computational feasibility, a random 5% subset of the dataset was selected using a fixed seed for reproducibility. The sampled dataset was then divided into training (80%), validation (10%), and test (10%) sets. This separation ensures that model selection and final evaluation are conducted fairly.

Key statistics include the total number of sessions, total number of hotel interactions, and the number of unique hotels. The average and median session lengths indicate that most sessions are relatively short, with a small number of longer sessions forming a long-tail distribution.

4 Exploratory Data Analysis

Analysis of session lengths reveals that most sessions are short, indicating that users typically explore only a limited number of

hotels. However, a small proportion of sessions are significantly longer, suggesting varied browsing behavior. This variability indicates that while most users interact with only a limited number of hotels, others engage in more extensive exploration. As shown in Figure 1, the distribution is highly skewed, with most sessions concentrated at lower lengths.

Hotel interactions follow a long-tail distribution, where a few hotels dominate interactions while the majority appear infrequently. This imbalance introduces challenges for recommendation models, particularly in handling rare items. Models must therefore balance accuracy for popular items with coverage of less frequent ones.

The sampled dataset retains similar session-length statistics to the original data, indicating that it preserves the overall session-length distribution. This supports its use for experimentation and model development, although it may not fully capture all aspects of the original dataset, such as the complete distribution of hotel frequencies.

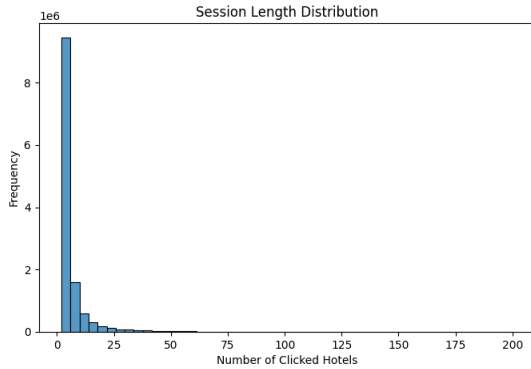


Figure 1: Distribution of session lengths in the dataset, showing that most sessions are short with a long tail of longer interactions.

5 Methodology

Each session is treated as a sequence of interactions, and the goal is to predict the next item based on preceding elements. Word2Vec is employed to learn dense vector representations of hotels by capturing co-occurrence patterns within sessions. This approach treats sessions as sentences and hotels as words, allowing the model to learn relationships based on contextual usage. Compared to more complex sequential models such as recurrent neural networks, Word2Vec provides a simpler and computationally efficient alternative, making it suitable for large-scale datasets. The resulting embeddings map hotels into a continuous space where similar items are positioned close to each other.

The model is implemented using the Gensim library with the skip-gram architecture ($sg = 1$) and negative sampling. The main hyperparameters include:

- $vector_size \in \{16, 32, 64\}$
- $window \in \{1, 2, 3, 4\}$
- $negative \in \{4, 8, 16, 32, 64\}$
- $alpha \in \{0.0001, 0.001, 0.01\}$
- minimum frequency threshold (min_count)
- number of training epochs

The experiments tuned vector size, window size, number of negative samples, and learning rate, while the minimum frequency threshold ($min_count = 5$) and the number of training epochs ($epochs = 10$) were kept fixed across all experiments. The window parameter controls the extent of local context captured within sessions, while the negative sampling parameter influences the model's ability to distinguish relevant from irrelevant items. The minimum frequency threshold plays a critical role in sparse datasets, as it removes infrequent hotels, improving training stability but potentially reducing coverage of rare items. These parameters collectively balance model complexity, training efficiency, and the ability to capture meaningful relationships.

Word2Vec operates by predicting surrounding items within a defined context window. In the skip-gram architecture, the model predicts neighboring hotels given a target hotel, encouraging items appearing in similar contexts to have similar embeddings. This design is particularly effective for improving representations of less frequent hotels. Negative sampling further enhances computational efficiency, making training feasible on large datasets. Additionally, embedding-based methods can generalize across similar items, allowing the model to recommend relevant hotels even if they have not been frequently observed together.

Three recommendation strategies are derived:

- Popularity-based model: ranks hotels by frequency
- Previous-item model: recommends items similar to the last interaction
- Context-based model: uses the average embedding of all prior interactions

Similarity between hotels is computed using the dot product of their embedding vectors without explicit normalization. While cosine similarity is commonly used in embedding-based systems, the dot product provides an effective and computationally efficient alternative for capturing relative similarity between items.

A minimum frequency threshold of $min_count = 5$ is applied during training, meaning that hotels appearing fewer than five times are excluded from the vocabulary. This improves training stability but reduces coverage of rare items.

The recommendation strategies apply different filtering rules. In the popularity-based and context-based models, all previously visited hotels in the session are excluded from recommendations to avoid trivial suggestions. In contrast, the previous-item model excludes only the reference hotel used for generating recommendations, allowing other previously seen hotels to still be recommended. This design reflects differences in how each method models user behavior.

6 Evaluation Strategy

Model performance is measured using Hits@10, which evaluates whether the correct next item appears within the top 10 recommendations. This metric reflects real-world scenarios where only a limited number of options are presented to users and is particularly suitable for ranking-based systems where exact prediction is less important than inclusion in top results.

For each session, all but the last interaction are used as input, while the final item serves as the target. Sessions lacking sufficient information or missing embeddings are excluded from evaluation. The number of evaluated and skipped sessions is reported to ensure transparency.

Although Hits@10 provides a clear measure of performance, it does not consider the exact ranking position of the correct item. Despite this limitation, it remains appropriate for evaluating recommendation quality in this context, as users typically interact with only a small number of recommendations.

It is important to note that sessions for which embeddings were not available (due to infrequent items filtered during training) were excluded from evaluation. As a result, the reported performance reflects only a subset of the data and may be slightly optimistic. This limitation is inherent to embedding-based approaches when applied to highly sparse datasets.

Model selection was initially performed using validation results obtained before hyperparameter tuning, comparing the previous-item and context-based approaches. However, hyperparameter tuning was subsequently conducted separately for each model, resulting in different optimized configurations.

After tuning, both models were retrained using the combined training and validation data and evaluated on the test set. Therefore, the final reported results represent a comparison between the best-performing configurations of each model, rather than a strictly validation-selected single model evaluated once on the test set.

This approach allows for a fair comparison of model capabilities but should be interpreted as a comparative evaluation rather than a strict model selection pipeline.

7 Experimental Procedure

The experiment follows a structured workflow. First, the Word2Vec model is trained on the training set using different hyperparameter configurations. A one-parameter-at-a-time approach is applied to identify optimal values while keeping other parameters fixed. This simplifies the tuning process while maintaining systematic evaluation.

Hyperparameter tuning was performed separately for the previous-item and context-based models using a one-parameter-at-a-time approach. For each model, parameters such as vector size, window size, number of negative samples, and learning rate were varied while keeping the remaining parameters fixed.

The final configurations correspond to the best-performing settings for each model individually, rather than a single shared configuration across all models.

For the previous-item model, the best-performing configuration included vector size = 64, window = 3, negative sampling = 8, and learning rate = 0.01.

Empirical observations indicated that moderate window sizes performed best, suggesting that short-range dependencies dominate user behavior. Larger context windows did not significantly improve performance, likely due to the dilution of relevant signals. Multiple configurations are evaluated under consistent conditions, allowing for systematic comparison across different parameter settings. In addition to quantitative evaluation, the learned

embeddings encode relationships between hotels based on co-occurrence patterns within sessions.

To further analyze the impact of individual hyperparameters, Figure 2 presents the variation of Hits@10 as each parameter is changed while keeping others fixed. The results show that performance improves with larger embedding sizes, reaching its best value at vector_size = 64. A window size of 3 provides the optimal context, while larger windows introduce noise. For negative sampling, the best performance is achieved at 8 samples. The learning rate has the strongest effect, with alpha = 0.01 significantly improving performance. These observations guided the selection of the final configuration for the previous-item model.

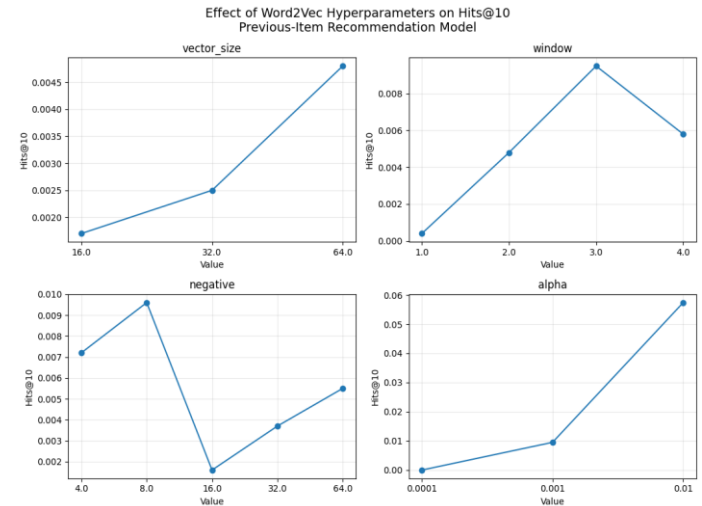


Figure 2: Effect of Word2Vec hyperparameters on Hits@10 for the previous-item recommendation model.

8 Results

The comparison of models using Hits@10 shows a clear performance gap:

Model	Hits@10
Popular Baseline	0.0035
Word2Vec Previous-Item	0.0592
Word2Vec Context-Based	0.0405

Several sessions were excluded due to missing embeddings, as discussed in the evaluation section, which should be considered when interpreting the results.

Embedding-based methods outperform the baseline, demonstrating the benefit of learning relationships between hotels. The previous-item model achieves the highest accuracy, indicating that recent interactions provide the most informative signal for prediction.

This suggests that local sequential patterns play a more important role than global session context in this dataset.

Furthermore, the relatively low absolute values are expected due to the extremely large item space, which contains a very large number of unique hotels. The sparsity of the dataset further contributes to the difficulty of the task, as many hotels appear infrequently, limiting the model’s ability to learn reliable representations. In such settings, even modest improvements over the baseline represent meaningful gains, as a random recommendation strategy would yield performance close to zero.

9 Discussion

The results emphasize the importance of sequential patterns in user behavior. The poor performance of the popularity baseline reflects its inability to capture session dynamics, while embedding-based approaches effectively model relationships between items.

The superior performance of the previous-item model indicates that user decisions are strongly influenced by their most recent interaction. In contrast, the context-based model, although incorporating more information, appears less effective due to the dilution of relevant signals when averaging embeddings.

Another contributing factor is the trade-off between simplicity and expressiveness, where simpler models capture strong local signals, while more complex representations may introduce noise.

Despite these improvements, overall performance remains limited due to data sparsity and the absence of additional contextual features, illustrating the inherent complexity of recommendation problems at scale.

This underscores the importance of well-structured session data in practical recommendation settings.

Another important observation is that embedding-based methods inherently favor frequently occurring items, as they receive more training signals. This bias may contribute to reduced performance on rare hotels, highlighting a trade-off between accuracy and coverage in sparse recommendation settings.

10 Conclusion

This study demonstrates that Word2Vec can effectively model session-based hotel recommendation by capturing relationships between items through sequential interactions. The results highlight that recent user interactions provide the strongest predictive signal, reinforcing the importance of local context in user behavior.

However, further improvements are needed to address data sparsity and incorporate richer contextual information. These limitations suggest that while the current approach provides a strong baseline, more advanced models and additional features are required to achieve higher performance.

Despite its simplicity, the previous-item Word2Vec model indicates that strong performance can be achieved by effectively leveraging local sequential patterns, suggesting that model complexity is not always the primary driver of recommendation quality.

11 Future Work and Improvements

Future improvements could involve more advanced sequential models such as GRU-based architectures or transformer-based approaches, which can capture long-range dependencies and model complex user behavior. These models have the potential to better represent evolving user behavior and improve recommendation performance by capturing richer interaction patterns within sessions.

In addition, incorporating multiple evaluation metrics such as Recall@K, Mean Reciprocal Rank (MRR), and Normalized Discounted Cumulative Gain (NDCG) would provide a more comprehensive assessment of recommendation quality. Including side information such as hotel attributes, location, and pricing could further improve performance, particularly for less frequent items.

12 Limitations

The dataset exhibits strong sparsity, limiting the model’s ability to learn reliable representations for rare items. As a result, recommendations for infrequently visited hotels may be less accurate. Word2Vec also does not explicitly model long-range dependencies beyond a fixed context window, which restricts its ability to capture complex sequential patterns.

Additionally, the use of sampled data may affect how well results generalize to the full dataset, as it does not fully reflect the full distribution of the original dataset. The reliance on a single evaluation metric further limits the assessment of model performance, as it does not capture ranking quality in detail. These limitations highlight the need for more comprehensive modeling and evaluation strategies in future work.

Another limitation arises from the handling of out-of-vocabulary (OOV) items. Due to the minimum frequency threshold applied during training (`min_count = 5`), some hotels do not receive embeddings. As a result, sessions containing such items are excluded from evaluation. This may introduce a slight upward bias in the reported performance, as the evaluation focuses only on sessions where the model has sufficient information to make predictions.

13 References

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